

Entrepreneurial Finance & Venture Capital

Session 2 · Lecture Notes | Free Cash Flow & the 3-Stage Model

1. Session Overview

Session 2 builds on Session 1's Strava discussion to develop a **rigorous but practical framework for valuing startups**. The core idea: instead of projecting every line item in a DCF, compress the whole model down to three intuitive drivers — profitability, asset intensity, and growth — and use those to populate a 3-stage free cash flow model.

- Why not a standard DCF? High-growth startups have no history, negative profits, and growth rates that would explode a terminal-value calculation.
- The 3 drivers (p , a , g) are *observable* from industry benchmarks even when firm-specific data is thin.
- Key insight carried forward from Session 1: cash flow $\neq$ profit, and the gap is largest for high-growth firms.

2. The Lemonade Stand — Cash Flow Intuition

The professor uses three versions of a lemonade stand (£100 cost, £150 revenue/day) to show how **identical profits can produce wildly different cash flows** depending on the business model.

Model	Description	Day-1 Cash	Month-end Cash	External Finance Needed
Cash	Customer pays immediately	£50	£1,500	£100 (start-up only)
Credit / Receivables	Customer pays at month-end (30-day tab)	−£100	−£3,000 then +£4,500	£3,000 (or £30,000 for 10 stands)
Subscription	Customer pays monthly in advance	£4,400	£1,500	Zero (customers finance growth)

*All three businesses have **identical profit (£50/day)**. The P&L tells you nothing about which is more valuable. Only the cash flow statement — which captures balance-sheet movements — reveals the difference.*

Key lessons from the lemonade stand:

- Asset intensity is the enemy of cash generation. Receivables lock up cash in assets rather than returning it as cash.
- Subscription / prepayment models produce **negative asset intensity**: customers finance you, so more growth = more cash (opposite of manufacturing).
- The subscription machine is worth the most — investors rightly pay higher multiples because the underlying cash-generating quality is superior.

3. Free Cash Flow Formula & the Three Drivers

Standard FCF (revision)

Starting from first principles:

$$\text{FCF} = \text{EBIAT} - \Delta \text{Net Fixed Assets} - \Delta \text{NOWC}$$

where $\text{EBIAT} = \text{EBIT} \times (1 - \text{tax rate})$, $\Delta \text{Net Fixed Assets} = \text{CapEx} - \text{Depreciation}$, $\Delta \text{NOWC} = \text{change in Net Operating Working Capital}$.

Why exclude financing (interest, dividends)? We want to measure the pure cash-generating quality of the operating machine, independent of how it is financed. Modigliani-Miller: capital structure doesn't change operating value (to a first approximation).

Compressed Formula: p, a, g

With the simplifying assumption that profitability (p) and asset intensity (a) stay constant within each stage, the formula collapses to:

$$\text{FCF}_t = \text{Sales}_t \times [p - a \times g / (1 + g)]$$

p (profitability)	EBIT / Sales in a given year. A margin measure — how much operating profit the business earns per £1 of revenue.
a (asset intensity)	Net Operating Assets / Sales. How much in operating assets (fixed + working capital) must be held per £1 of revenue. Positive = asset-heavy; negative = customers pre-pay (subscription).
g (growth)	Year-on-year growth rate of sales. Interacts with asset intensity: if $a > 0$ and $g > 0$, you need to keep investing in assets to support growth, which drains cash.

How the Three Drivers Interact

Driver change	Effect on FCF	Condition
p ↑	FCF ↑ (unambiguously)	Always
a ↑	FCF ↓	When $g > 0$
a ↓ (goes negative)	FCF ↑ (can exceed profit)	When $g > 0$
g ↑	FCF ↓	When $a > 0$ (asset-heavy)
g ↑	FCF ↑	When $a < 0$ (subscription/prepay)
g → 0	FCF → profit	Always (at zero growth, $\text{FCF} = \text{EBIAT}$)

Critical insight — when are profit multiples valid? When g is low (mature companies), $\text{FCF} \approx \text{profit}$, so EBITDA multiples are a reasonable proxy. For high-growth startups (g is huge), profit multiples are misleading — you must model cash flows explicitly.

4. Why Standard DCF Fails for Startups

No history	Cannot project from past financials; the company may have no revenue yet.
Negative EBIT	FCF formula requires EBIAT; negative profits make the starting number meaningless.
Explosive growth	A 20% month-on-month growth rate plugged into a terminal value formula produces a number larger than world GDP. The terminal-value calculation only converges when $g < \text{discount rate}$.
No observable beta	These are private companies, so no market data exists to estimate systematic risk. Convention: use ~10% discount rate ($\beta \leq 2$, typical risk-free rate).
Power-law outcomes	Sensitivity analysis around a single DCF scenario assumes near-normal distributions. Reality: 90%+ chance of zero, tiny chance of massive return.

***The solution:** the 3-stage model handles explosive early growth explicitly, allows it to slow in a middle phase, then reverts to a stable terminal growth rate. The p-a-g parameterisation keeps the model tractable even with sparse data.*

5. The 3-Stage Free Cash Flow Model

The model extends a standard DCF by using **three explicit phases** rather than projecting year-by-year and then jumping to a single terminal value.

Stage	Years (typical)	Growth (g)	Profitability (p)	Asset Intensity (a)
Stage 1 Explosive growth	Years 1–5	Very high (e.g., 50%/yr)	Negative or low	Often negative (subscription) or high (hardware)
Stage 2 Scaling	Years 6–10	Moderating (e.g., 35%/yr)	Rising as scale kicks in	Converging toward industry norm
Stage 3 Terminal	Year 11+	Low, stable (~3%)	Mature industry level	Stable industry level

Inputs Required

Starting conditions: Sales_0 , EBIT_0 , $\text{Net Operating Assets}_0$. Then for each stage, three parameters:

- Growth rate (g)
- Profitability margin ($p = \text{EBIT} / \text{Sales}$)
- Asset intensity ($a = \text{Net Operating Assets} / \text{Sales}$)

That's 9 forward-looking numbers plus 3 starting conditions — tractable even for a company with no history.

Worked Example from Lecture

Starting company: £10M sales, –20% profit margin, asset intensity +10%.

- Stage 1: $g = 50\%/yr$, p rising from -20% to -5% , $a = +10\%$
- Stage 2: $g = 35\%/yr$, $p = +2\%$, $a = +10\%$
- Terminal: $g = 3\%$, stable margins

Result 1 — Asset-heavy ($a = +10\%$):

- After 10 years: $\sim£340M$ revenue but $FCF = -£2M/yr$ in Stage 2.
- Cumulative external finance needed to reach Year 10: $\sim£22.5M$.
- Valuation (10% discount rate, conditional on the growth path): $\sim£173M$.

Result 2 — Subscription model ($a = -10\%$):

- Same revenue and profit trajectory.
- FCF turns strongly positive: cumulative $FCF = +£45M$ (no external finance needed).
- Valuation: $\sim£219M$.

Takeaway: flipping asset intensity from $+10\%$ to -10% adds $\sim£46M$ in value and eliminates the need for external financing — all else equal. This is why subscription SaaS companies command premium multiples.

6. Profitability vs Asset Intensity Trade-offs

The three drivers are not independent — operational decisions move them in linked ways.

Raise prices	$p \uparrow$ but $g \downarrow$ (fewer customers). A trade-off, not a free lunch.
Offer discount for early payment	$a \downarrow$ (cash in sooner) but $p \downarrow$ (margin given up). Worth it when cost of external finance $>$ cost of the discount.
Outsource manufacturing	$a \downarrow$ (no capex on your balance sheet) but $p \downarrow$ (supplier earns a margin on top of costs). Trade-off between balance-sheet health and income-statement margins.
Network effects	Reduce CAC over time ($p \uparrow$) and reduce churn (more stable g). The best kind of operational leverage — improves multiple drivers simultaneously.

7. Connecting to Startup Valuation

Two Ways to Handle Uncertainty

Approach A DCF with probability weights	Model the 'success scenario' using the 3-stage model. Multiply the resulting valuation by the (small) probability of that path. Add ~ 0 for all the failure paths. E.g., $5\% \times £200M = £10M$ today.
Approach B VC method with high discount rate	Assume the company reaches a target exit value (e.g., $4-5\times$ revenue in Year 5). Discount back using a very high rate (50–70%) to reflect execution risk. The high rate is a shorthand for the probability of failure.

The two approaches are conceptually equivalent — they embed the same probability-of-success belief, just packaged differently. The VC method is faster; the DCF approach is more transparent about assumptions.

Revenue Multiples in SaaS — Why They Work

Vcs often value SaaS companies on a multiple of revenue rather than profit. This is rational when:

- The company is growing fast (g is high) so profit is artificially depressed by investment.
- Revenue growth is a leading indicator of future cash flow for subscription businesses (because negative asset intensity means high growth $\rightarrow$ high FCF in future).
- The revenue multiple adjusts for growth rate: faster-growing companies get higher multiples because the FCF formula shows higher value when g is large and a is negative.

Warning: *blindly applying VC/SaaS multiples to capital-intensive industries is a well-documented mistake. After the dot-com era, solar companies were briefly valued like internet companies — wrong, because a , p , and g profiles are completely different.*

8. Multi-Stage Financing & Adverse Selection

The lecture closes with a discussion of **why VCs invest in stages** rather than all at once — and what that means for early-stage investors.

Why stage investments?	In public markets, diversification is the risk-management tool. In private markets, multi-stage financing plays an analogous role: invest a little now, observe new information, then decide whether to continue. Each round is a real option.
Information improves over time	Later rounds are less mispriced because more data exists. But they are also more competitive — top VCs (Sequoia etc.) crowd out early investors.
Adverse selection	If you are the only bidder on a deal, ask: why did everyone else pass? Being oversubscribed is a positive signal; having 100% allocation is a warning sign.
Angel investor vulnerability	Early investors take the most risk but often lack funds to follow on. Later VCs can threaten to walk unless early investors accept dilution. Angels must commit to the game of chicken — or be squeezed out.

9. Key Takeaways

Business model determines cash flow	Same profit, completely different cash flows. Subscription > credit > cash for valuation purposes.
$FCF = Sales \times [p - a \cdot g / (1+g)]$	The three-driver formula. Memorise it and understand each lever.
Asset intensity $\times$ growth = cash drain	For positive- a businesses, fast growth consumes cash. For negative- a (subscription), fast growth generates cash — the counter-intuitive result we saw with Strava in Session 1.

Profit $\approx$ FCF only when $g \approx 0$	EBITDA multiples are fine for mature companies; dangerous for startups.
3-stage model	Stage 1 (explosive, often loss-making) $\rightarrow$ Stage 2 (scaling, margins improve) $\rightarrow$ Stage 3 (terminal, stable $g \sim 3\%$). Just 9 parameters needed.
Two valuation methods	DCF with probability weights OR VC method with high discount rate. Same underlying logic; different presentation.
Adverse selection	In private markets, always ask why a deal is available to you. The best deals are oversubscribed; the worst are unclaimed.

Next session (Session 3): VC valuation method in depth — IRR targets, dilution, cap tables, and staged financing mechanics.